Java String Programs

50 Practice Questions with Sample Test Cases

FOR STUDENT PRACTICE — NO CODE INCLUDED

Predict the output • Write the program • Verify against test cases

Java String Programs

Practice Questions 1–25 (with Sample Test Cases)

FOR STUDENT PRACTICE — NO CODE INCLUDED

Predict the output • Write the program • Verify against test cases

SEARCHING & LOCATING CHARACTERS

1 Find First Occurrence of a Character

Find the index of the first occurrence of a given character in a string.

```
"programming", 'g' → 3
"hello", 'l' → 2
"banana", 'a' → 1
```

2 Find Last Occurrence of a Character

Find the index of the last occurrence of a given character in a string.

```
"programming", 'g' → 10
"banana", 'a' → 5
"hello", 'l' → 3
```

3 Search a Character Without Using indexOf()

Check whether a character exists in a string without using the built-in indexOf() method.

```
"welcome", 'c' → Found at index 3
"java", 'v' → Found at index 2
"test", 'x' → Not Found
```

4 Search a Character (Not Present)

Search for a character and correctly report when it does not exist in the string.

```
"apple", 'z' → Character not found
"orange", 'q' → Character not found
"mango", 'm' → Found at index 0
```

5 Search a Substring

Check whether a given substring exists within a string, and print its starting index.

```
"hello world", "world" → Found at index 6
"java programming", "gram" → Found at index 8
"computer", "xyz" → Not Found
```

COUNTING OCCURRENCES

6 Count Occurrence of a Character

Count how many times a given character appears in a string.

```
"banana", 'a' → 3
"mississippi", 's' → 4
"hello", 'z' → 0
```

7 Count Occurrence of a Word

Count how many times a given word appears in a sentence.

```
"the cat sat on the mat the cat ran", "the" → 3
"good morning good night", "good" → 2
"one two three", "four" → 0
```

8 Find All Positions of a Character

Print all index positions where a given character appears in a string.

```
"banana", 'a' → [1, 3, 5]
"programming", 'r' → [1, 4]
"hello", 'l' → [2, 3]
```

STRING MATCHING METHODS

9 Check startsWith()

Check whether a string starts with a given prefix.

```
"Hello World", "Hello" → true
"programming", "pro" → true
"java", "script" → false
```

10 Check endsWith()

Check whether a string ends with a given suffix.

```
"testing.java", ".java" → true
"hello world", "world" → true
"laptop", "top10" → false
```

11 Check contains()

Check whether a string contains a given substring anywhere within it.

```
"the quick brown fox", "brown" → true
"database", "base" → true
"network", "internet" → false
```

INDEX-BASED LOOKUPS

12 Find Index of Every Vowel

Print the index positions of every vowel present in a string.

```
"education" → [0, 2, 4, 6, 8]
"hello" → [1, 4]
"sky" → [] (no vowels)
```

13 Find Index of Every Digit

Print the index positions of every digit present in a string.

```
"a1b2c3" → [1, 3, 5]
"user123name" → [4, 5, 6]
"hello" → [] (no digits)
```

CHARACTER TYPE COUNTING

14 Count Uppercase Letters

Count the number of uppercase letters in a string.

```
"Hello World" → 2
"JAVA Programming" → 5
"no caps here" → 0
```

15 Count Lowercase Letters

Count the number of lowercase letters in a string.

```
"Hello World" → 8
"ALLCAPS" → 0
"MixedCASE" → 5
```

16 Count Digits

Count the number of digit characters in a string.

```
"abc123xyz456" → 6
"no digits here" → 0
"room007" → 3
```

17 Count Vowels

Count the number of vowels (a, e, i, o, u) in a string.

```
"education" → 5
"programming" → 3
"sky" → 0
```

18 Count Consonants

Count the number of consonants in a string.

```
"education" → 4
"hello" → 3
"aeiou" → 0
```

19 Count Spaces

Count the number of space characters in a string.

```
"the sky is blue" → 3
"nospaceshere" → 0
"a b c d e" → 4
```

20 Count Special Characters

Count non-alphanumeric characters (excluding spaces) in a string.

```
"hello@world#2026!" → 3
"price: $100.50" → 3
"plaintext" → 0
```

WORD & CHARACTER ANALYSIS

21 Count Words

Count the number of words in a sentence.

```
"Java is a powerful language" → 5
"single" → 1
" multiple spaces between words " → 3
```

22 Count Alphabet Characters

Count the number of alphabetic characters (a–z, A–Z) in a string.

```
"Room 007 is ready!" → 12
"12345" → 0
"Hello World" → 10
```

23 Count Numeric Characters

Count the number of numeric (digit) characters in a string.

```
"Order#2024 placed on 15-08" → 8
"no numbers" → 0
"a1b2c3d4" → 4
```

24 Count Occurrence of Each Character

Display the frequency count of every distinct character in a string.

```
"banana" → b=1, a=3, n=2
"hello" → h=1, e=1, l=2, o=1
"aabbcc" → a=2, b=2, c=2
```

25 Count Frequency of Every Word

Display the frequency count of every distinct word in a sentence.

```
"cat dog cat bird dog cat" → cat=3, dog=2, bird=1
"hi hi hi" → hi=3
"one two three" → one=1, two=1, three=1
```